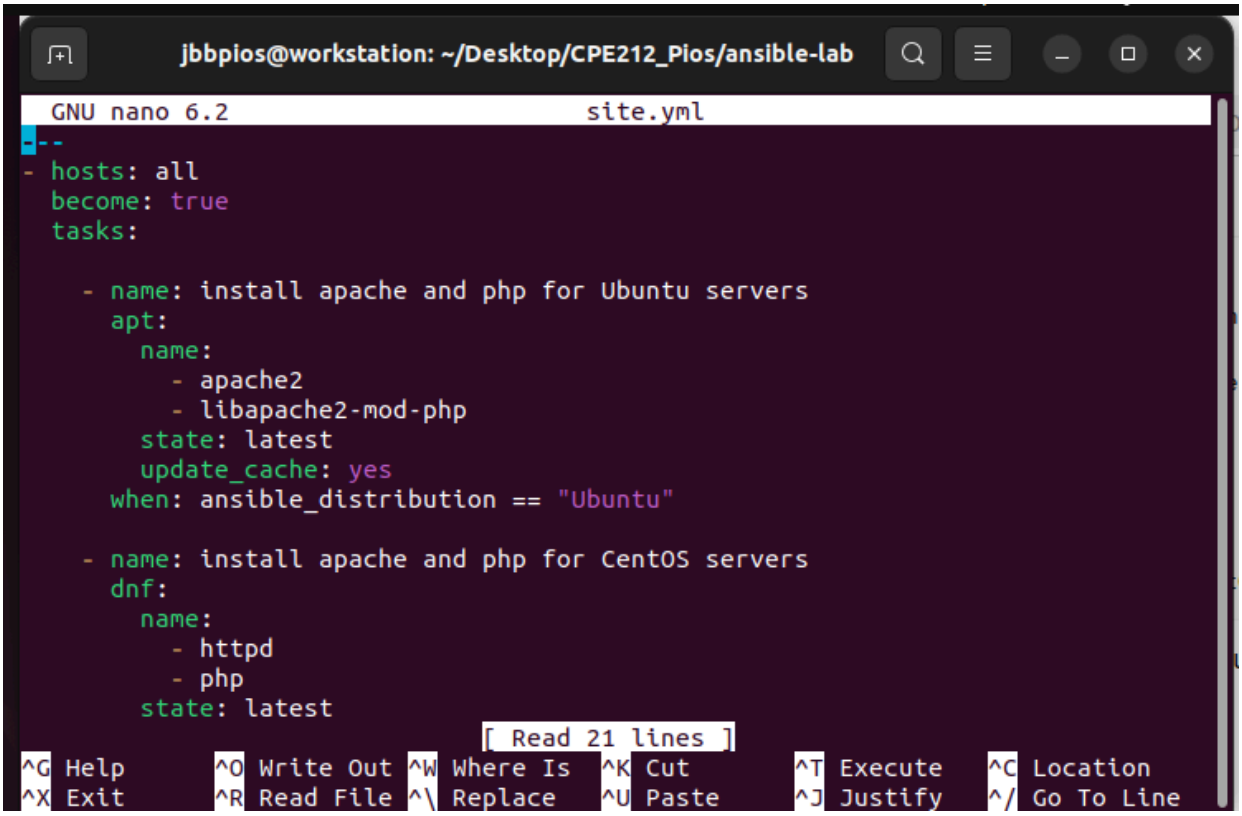
COMPUTER ENGINEERING

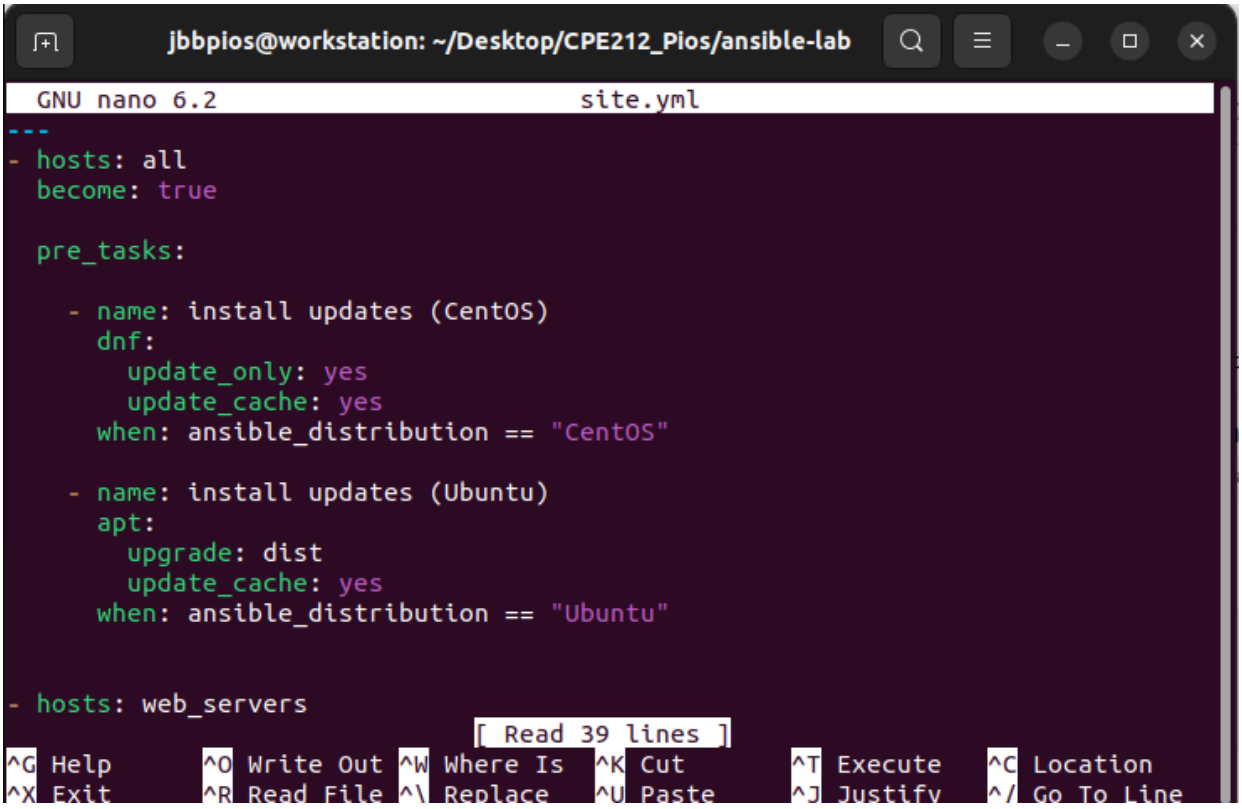
QUEZON CITY

COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	September 05, 2026	**DATE SUBMITTED:**	September 05, 2026
NAME:	Pios, Joshua Paul B.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

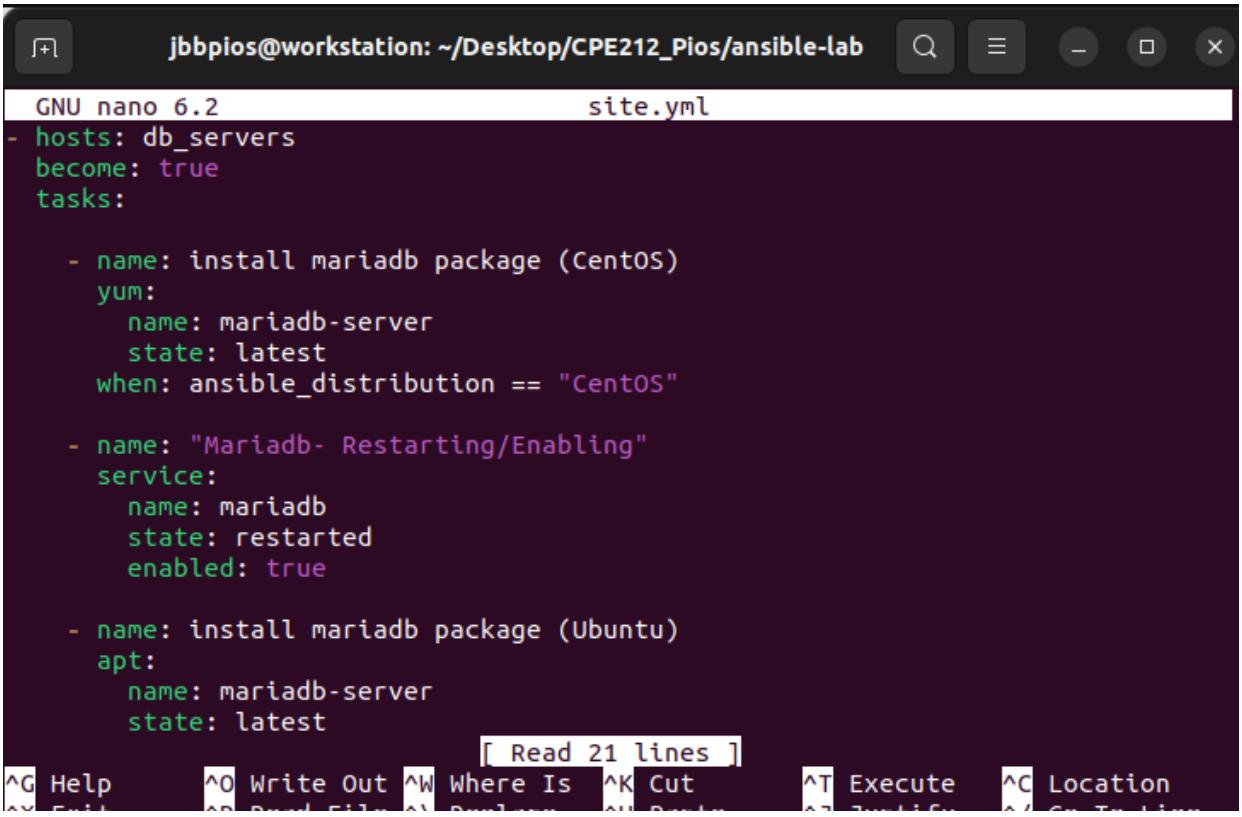
ACTIVITY NO. 6.1 **Targeting Specific Nodes**

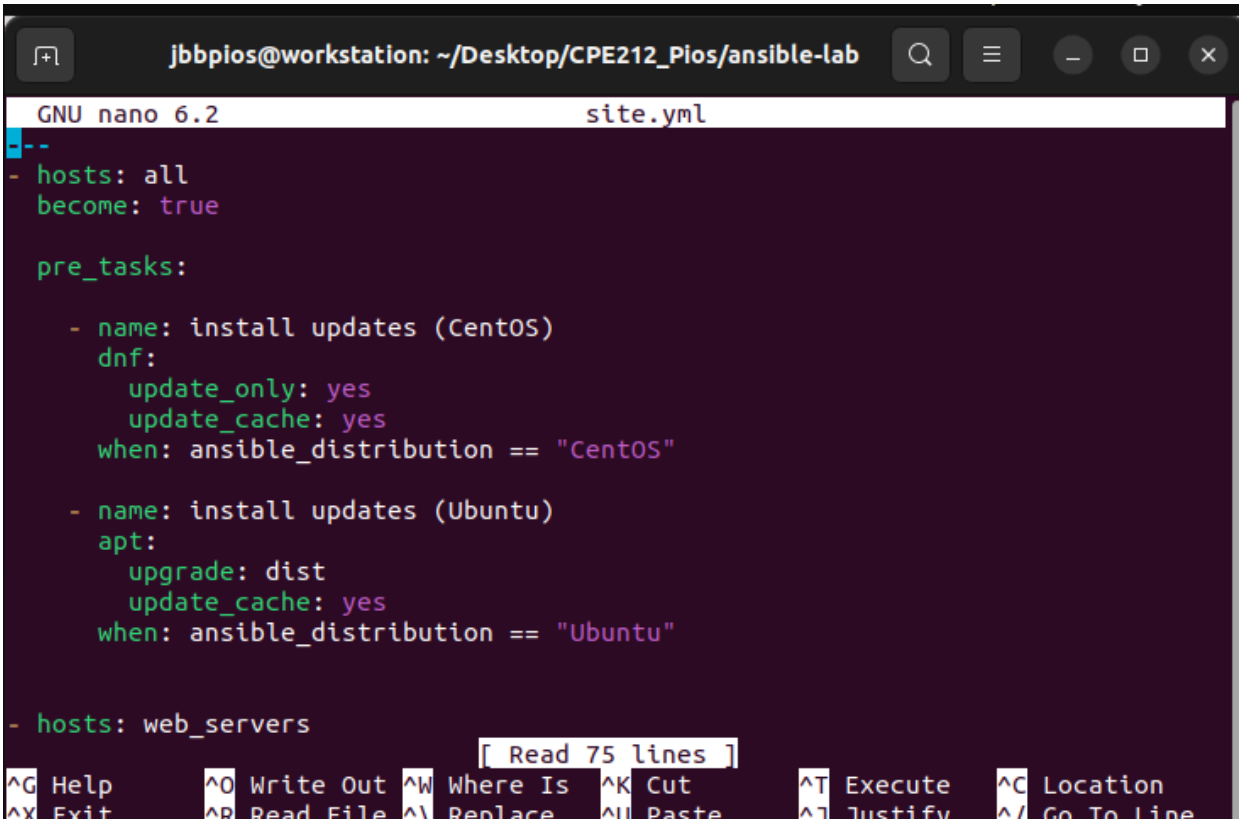
1. PROCEDURE OUTPUT



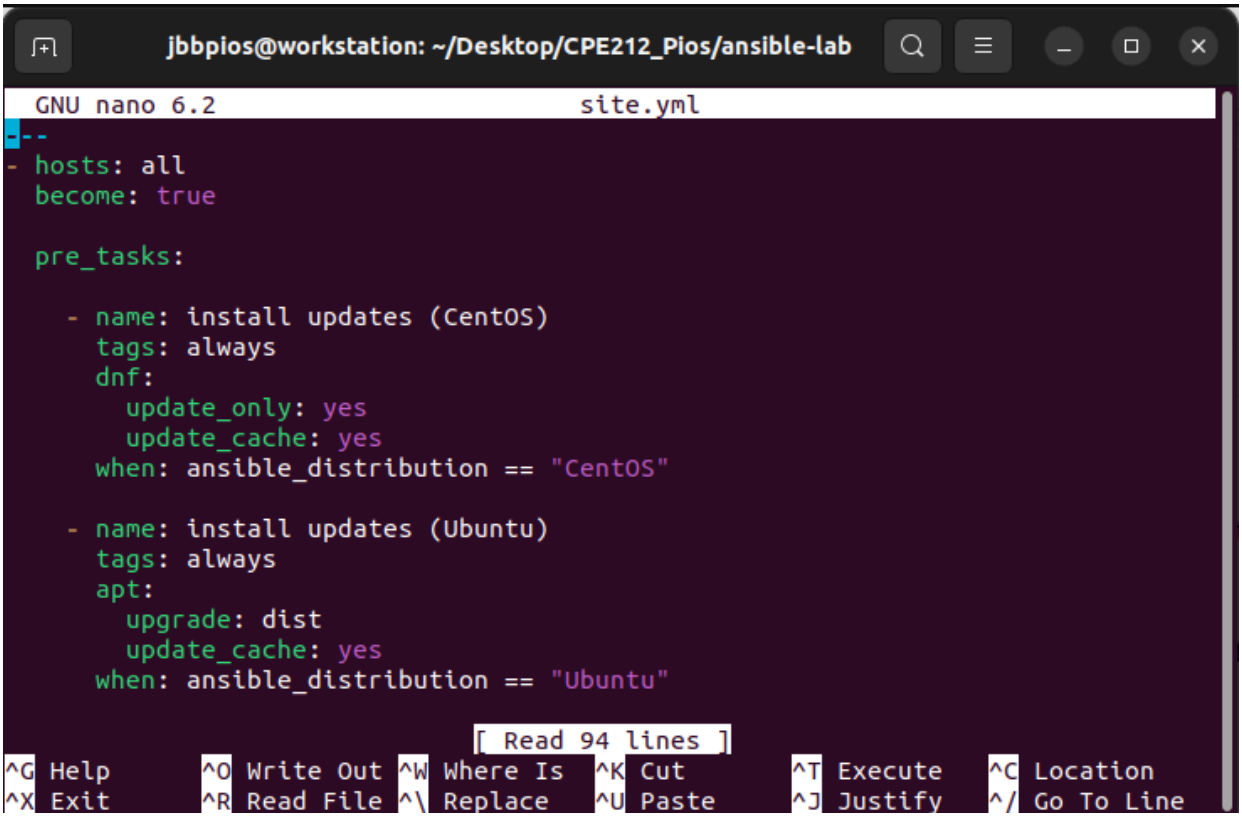

jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab
GNU nano 6.2 site.yml

- hosts: all
become: true
pre_tasks:
- name: install updates (CentOS)
dnf:
update_only: yes
update_cache: yes
when: ansible_distribution == "CentOS"
- name: install updates (Ubuntu)
apt:
upgrade: dist
update_cache: yes
when: ansible_distribution == "Ubuntu"
- hosts: web_servers
[Read 39 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

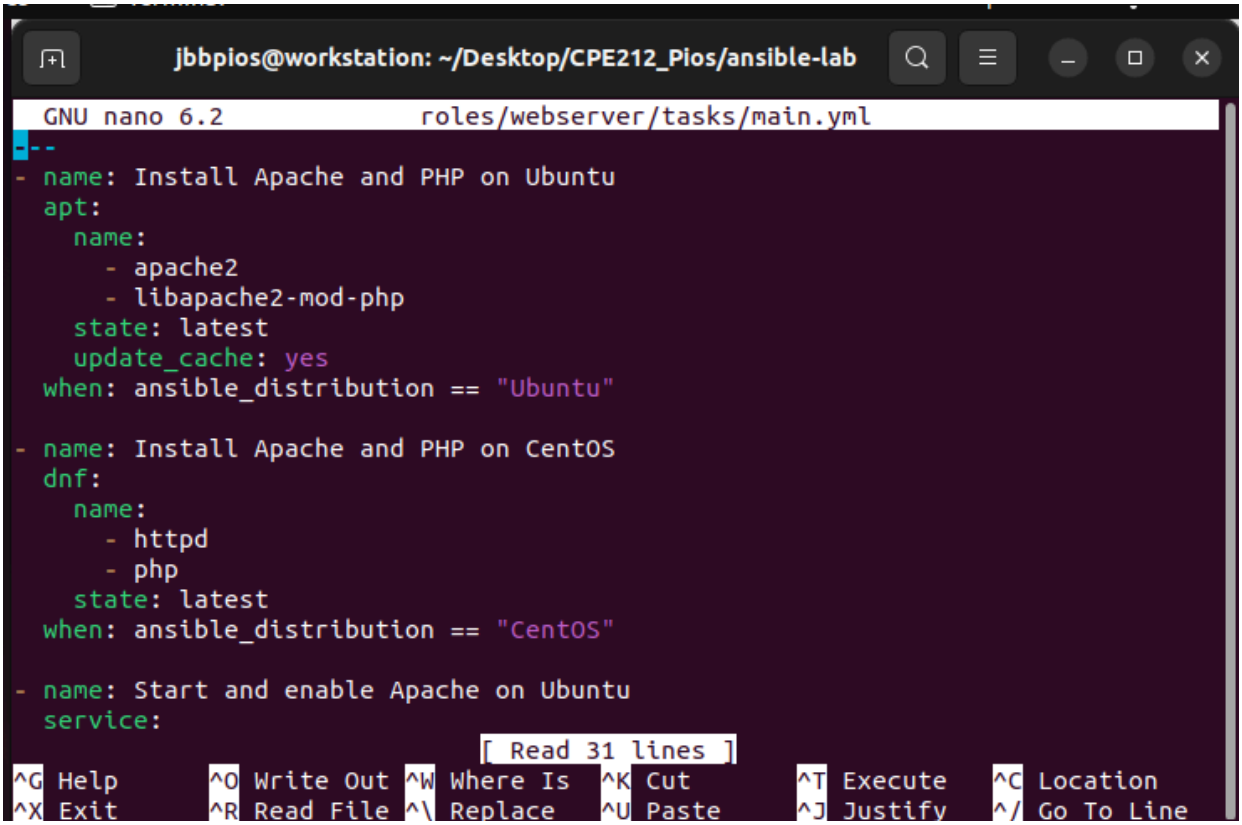

jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab
GNU nano 6.2 site.yml
- hosts: db_servers
become: true
tasks:
- name: install mariadb package (CentOS)
yum:
name: mariadb-server
state: latest
when: ansible_distribution == "CentOS"
- name: "Mariadb- Restarting/Enabling"
service:
name: mariadb
state: restarted
enabled: true
- name: install mariadb package (Ubuntu)
apt:
name: mariadb-server
state: latest
[Read 21 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location


jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab
GNU nano 6.2 site.yml

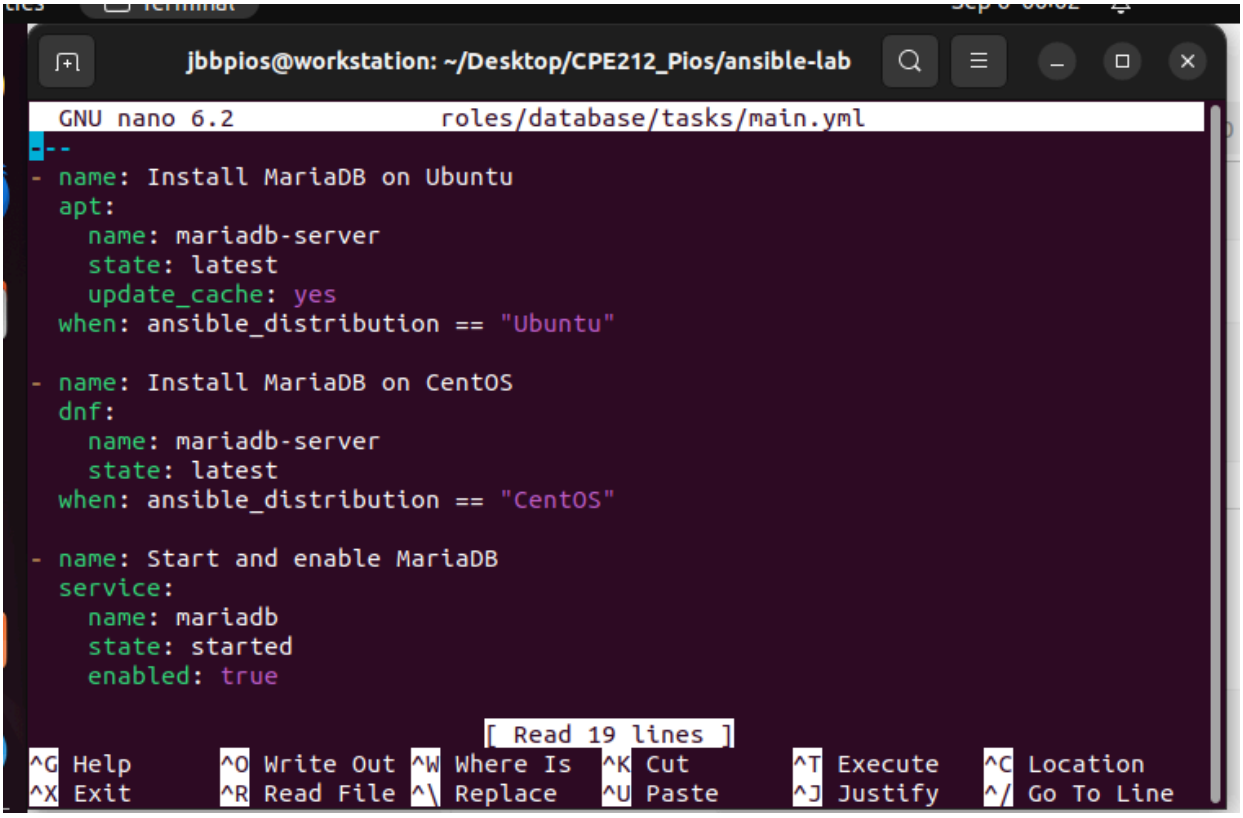
- hosts: all
become: true
pre_tasks:
- name: install updates (CentOS)
dnf:
update_only: yes
update_cache: yes
when: ansible_distribution == "CentOS"
- name: install updates (Ubuntu)
apt:
upgrade: dist
update_cache: yes
when: ansible_distribution == "Ubuntu"
- hosts: web_servers
[Read 75 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line



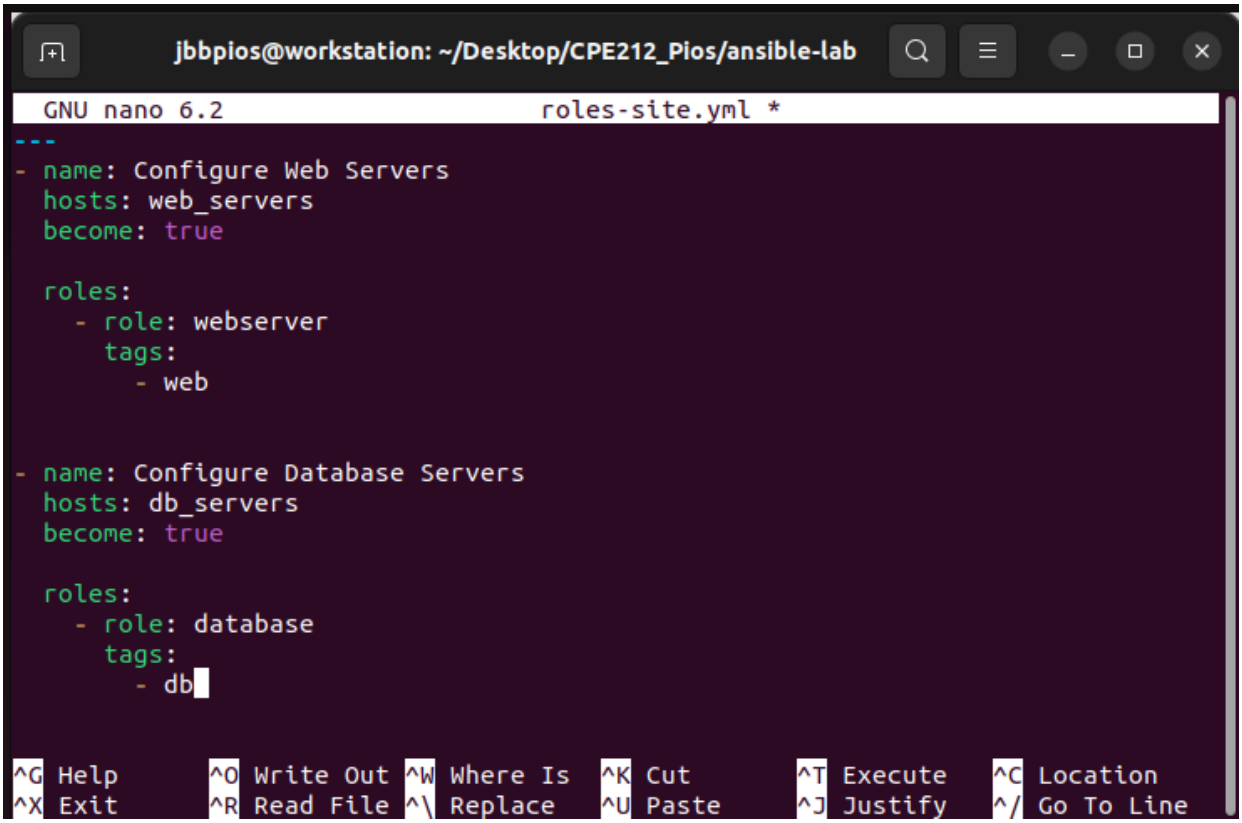
2. ANSWERS TO SUPPLEMENTAL ACTIVITY


jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab
GNU nano 6.2 roles/webserver/tasks/main.yml

- name: Install Apache and PHP on Ubuntu
apt:
name:
- apache2
- libapache2-mod-php
state: latest
update_cache: yes
when: ansible_distribution == "Ubuntu"
- name: Install Apache and PHP on CentOS
dnf:
name:
- httpd
- php
state: latest
when: ansible_distribution == "CentOS"
- name: Start and enable Apache on Ubuntu
service:
[Read 31 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


jbbpios@workstation: ~/Desktop/CPE212_Pios/ansible-lab
GNU nano 6.2 roles/database/tasks/main.yml

- name: Install MariaDB on Ubuntu
apt:
name: mariadb-server
state: latest
update_cache: yes
when: ansible_distribution == "Ubuntu"
- name: Install MariaDB on CentOS
dnf:
name: mariadb-server
state: latest
when: ansible_distribution == "CentOS"
- name: Start and enable MariaDB
service:
name: mariadb
state: started
enabled: true
[Read 19 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line



1. What is the importance of putting our remote servers into groups?
Putting remote servers into groups allows Ansible to organize machines according to their roles, such as web, database, and file servers. This makes playbooks easier to manage because tasks can be applied only to the servers that need them instead of affecting every remote host.

2. What is the importance of tags in playbooks?
Tags allow specific tasks or sections of a playbook to be executed without running the entire playbook. They make automation more flexible and efficient, especially when only a particular service or configuration needs to be updated.

3. Why do you think some services need to be managed automatically in playbooks?
Some services do not automatically start after their packages are installed, so they need to be explicitly started and enabled through Ansible. Automating service management ensures that the required services are consistently running and reduces the need for manual configuration on every server.

Conclusion/Learnings:
This activity taught me how Ansible can target specific remote servers by organizing them into inventory groups based on their roles. I also learned how tags can be used to selectively execute tasks and how services such as Apache and MariaDB can be managed automatically through Ansible. Overall, the activity demonstrated how grouping, tagging, and service management make server administration more organized, efficient, and consistent.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic

work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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